1 Mechanical Structure Module

Pre-Lab: 1.3.1.a

1.1 Module Overview

The primary output of this module is a prototype of the pinball machine enclosure, a design plan for the intended layout of the playfield, and a testing platform for the plunger system.

In addition to the design, fabrication, and characterization of the functional mechanical structure of the pinball machine and the plunger system, this module also has other activities that build relevant skills both inside and outside of the lab sessions. These other activities include introductions to system design principles, iterative prototyping, CAD, laser cutting, and 3D printing.

Contents

1 Mechanical Structure Module 1
1.1 Module Overview 1
1.1.1 Learning goals 2
1.2 Deliverables 2
1.3 Activities 3
1.3.1 Mechanical Prototyping Tools: CAD, 3D printing and laser cutting 3
1.3.2 System-level design: Block diagrams, system requirements, and design specifications 8
1.3.3 Pinball System Design Plan 11
1.3.4 Pinball Machine Playfield Mechanical Design Cycle 11
1.3.5 Pinball Plunger Testbed Mechanical Design Cycle 12

1.1.1 Learning goals

Purposes ('What this module's activities seek to develop')

- Prepare students/teams for independent design project work
    - Systematic approach to prototyping and concept evaluation
    - Mechanical prototyping skills
    - Knowledge of available tools, materials, and methods
- Develop/practice a disciplined and systematic approach to the design process that includes iteration and directed prototyping with clear metrics and learning
    - Asking and answering questions with prototypes
    - Documenting design progression

Knowledge/Functional goals ("Students/teams will (be able to)…")

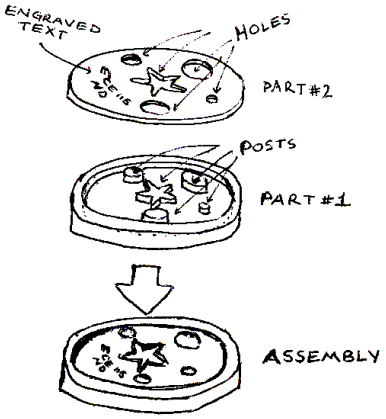
- Create a design based on system requirements
- Identify potential challenges/questions about a design and test them through prototyping and experiments
- Document and evaluate test results appropriately
- Create plans for next-step prototype(s)/experiments based on existing prototypes/experiments
- Create 3D drawings of parts and an assembly in CAD (SolidWorks)
- Use rapid prototyping tools, including hand tools, laser cutter and 3D printer, to create parts and assemblies using drawings and CAD files
- Know common practical limitations of rapid prototyping techniques, including resolution, tolerances, etc.
- Understanding and follow safe mechanical prototyping practices

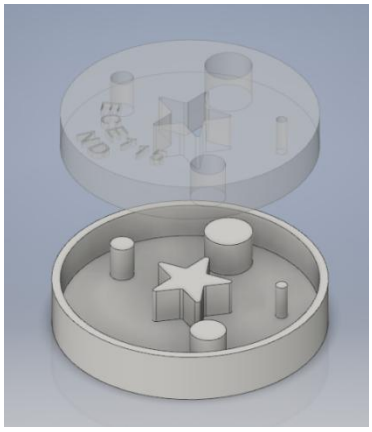
1.2 Deliverables

The list of deliverables for this module is uploaded in a document entitled Module 1 Deliverables on Canvas; **they are highlighted in green in this document**. Please create a **single .pdf report** and upload to to Canvas, and a zip file of the design files.

1.3 Activities

1.3.1 Mechanical Prototyping Tools: CAD, 3D printing and laser cutting





1.3.1.a CAD for 3D printing and laser cutting: Pre-lab

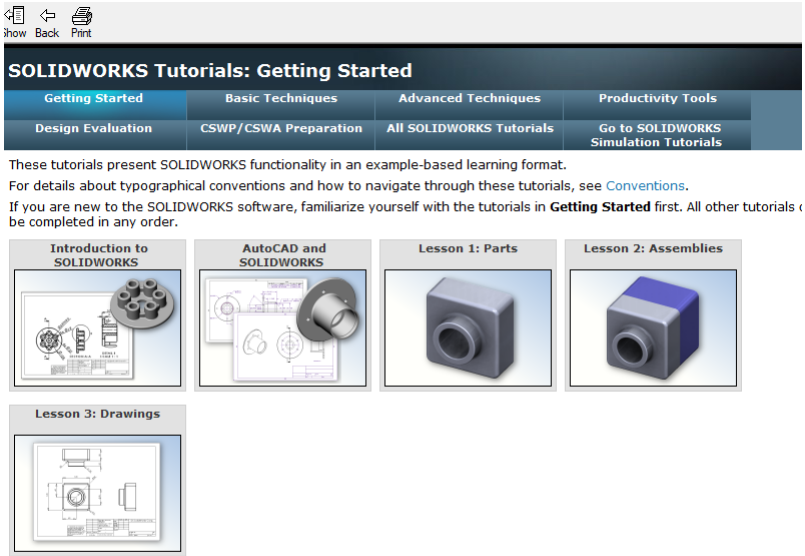
This part of the lab introduces CAD tools (in this case, SolidWorks) as a method for rapid prototyping.

The goal is for each student to produce a specific two-part assembly using SolidWorks that meets the stated requirements. Each student will prepare files for production of the two parts—one for the laser cutter and one for the 3D printer—according the specific instructions in the following steps.

IMPORTANT FOR MAC/LINUX USERS: We encourage students to use SolidWorks as it is the most sought after for industry, but we understand that it is only available on Windows. An alternative version is Autodesk Fusion 360 (online) and Onshape (online) which are newer and targets a more broadly skilled audience. Note that this is the only location where we will bring up these alternatives since the industry is overwhelmingly Windows/Solidworks. If you are stuck with using either Fusion/Onshape, any time you see mention of Solidworks you should simply assume that you should generate the same Fusion/Onshape-generated design file.

1.3.1.a.1 CAD for 3D printing and laser cutting: Pre-lab Steps

1. **If you have never used SolidWorks before, begin with a tutorial. The concepts to learn are how to (1) make Parts, Assemblies, and Drawings.** You may skip this section if you are already familiar with creating these. Note that you will submit parts, assemblies and drawings for grade so it is your responsibility to produce the properly defined parts and assemblies – with constraints, smart dimensions, and mates. If this was not immediately clear, then go through the five modules below (found by searching tutorials in Solidworks search bar).



1. Create an assembly, comprising two mating parts, according to the following specifications/diagrams. Export the assembly as "assembly_<yourname>_ece115_prelab1.sldasm" and save a screenshot of the assembly as "assembly_<yourname>_ece115_prelab1.jpg"
    a. Part #1 (we will manufacture this during the lab session using PLA plastic, via an FDM process)
        i. Cylindrical form factor, with outer diameter <=55mm
        ii. Has a central 'cutaway' region with uniform depth and diameter >= 41mm
        iii. Has a lip to the central 'cutaway' with minimum width of 2mm
        iv. 'Cutaway' has raised unique shape design, that has feature height flush with the top surface of the 'cutaway' lip (Figure 1 uses a star as an example)
        v. 'Cutaway' contains 3-5 cylindrical posts, each with different diameter, covering the range of 2mm-8mm
        vi. Has nominal material cost ~$5 or less (see 5a, below)
    b. Part #2 (we will laser cut this during the lab session out of ¼" extruded acrylic)
        i. Must mate with Part #1 (see Figure 1)
        ii. Has cut or etching that contains 'ECE115' and your initials
    c. The mated assembly of parts #1 and #2 will have a height of <=12mm”

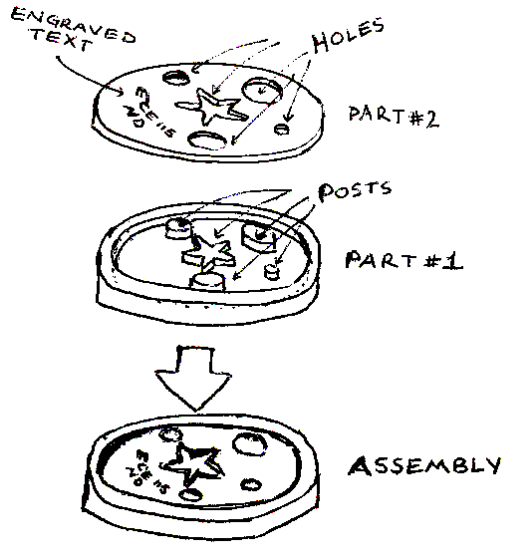
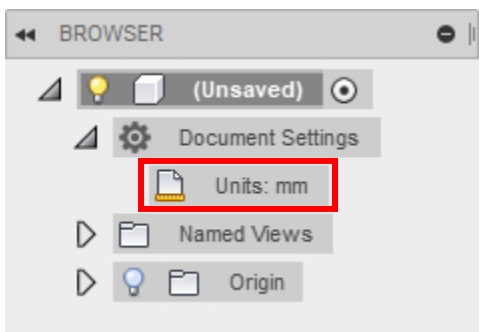


Figure 1: Rough sketches of Parts #1 and #2 (top), with assembly (bottom)

2. Export a .stl file of Part #1 from Step 3a. We will load this on the 3D printing software to make your part. **Ensure your units are set to mm prior to exporting, otherwise your scaling may be off.**



   a. Please name it according to the following: "part1_<yourname>_ece115_prelab1.stl"
3. Create a simple line drawing of Part #2 from Step 3b—see Figure 2 for an example. We will load this on the laser cutter to make your part. You will need to perform a couple of processing steps to set up the file for laser cutting. **Note: be sure that the size/scaling in the final .pdf is accurate.**
4. Save the layer/sketch as a .dxf. **Ensure your units are set to mm prior to saving.**
   a. Open the .dxf file in Inkscape.
   b. Remove any stray marks or words, center the drawing, and crop the document to fit the drawing (File > Document Properties > Resize Page to Content).
   c. Export simple line drawing as a .pdf: "part2_<yourname>_ece115_prelab1.pdf". Use the highest quality settings when saving your PDF.
   d. Finally, since the cutter uses line color and width to encode its cutting operation, edit the .pdf line drawing so that it has the following properties (Object > Fill and Stroke):
      i. The *only* lines or marks in the file are the cut/etch lines for the part (no header/footer/border/measurement lines, etc.)

ii. Cutting line color is true red (R,G,B = 255,0,0)
iii. Cutting line width is 0.02mm
iv. (optional) Etching line color is blue (R,G,B = 0,0,255), and etching line width is not important, so set it to ~0.1mm
v. File name: “part2_<yourname>_ece115_prelab1.pdf”

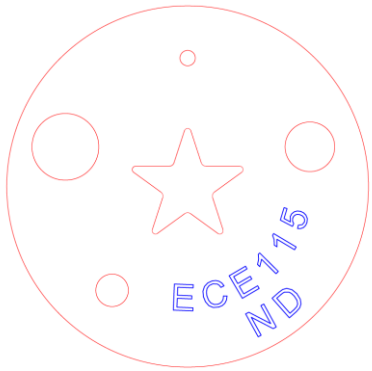


Figure 2: Example of a correctly-formatted .pdf file for laser cutting. Note that there are no stray marks and the lines are specific colors and thicknesses (per the instructions).

5. Answer the following questions about the above part designs:

In a table, outline as many differences as you can regarding mechanical strength, brittleness, sensitivity to humidity and temperature, and cost. between PLA, ABS, and PETG (materials used in FDM printers) and standard SLA resin. Compare as these materials to wood (birch) and aluminum. Use internet resources to determine this.

(societal impact) In another table, outline these materials based on them as waste products: biodegradability, environmental impact, recyclability, and compare them with wood and then aluminum.

1.3.1.b CAD, 3D printing and laser cutting: Lab work

In this part of the lab you will:

- Use measurement tools in the lab with your (or your teammate's) already-cut/printed parts
- Table how measurements of the parts compare with both the design specification from the pre-lab and with your part file(s) in SolidWorks, including your measurement accuracy (e.g. "+/-1mm")
- Compare the measurements with the CAD file(s). Where there are mismatches, please explain or hypothesize how to improve any discrepancies.

1.3.1.b.1 Laser cutting

In this part of the lab you will:

- Use your SolidWorks-generated and Inkscape-modified file from the pre-lab exercise (Part #2; .pdf file) to cut your part from ¼" cast acrylic using a laser engraver/cutter
- Learn about file format requirements first-hand
- Learn about the process and limitations of laser cutting acrylic

As the laser cutter requires files exactly in a particular format and character (as described in the pre-lab) so that it knows what kinds of cuts or marks to make on which lines. Please make the file with the correct scaling, line widths, and line color(s). If your 3D-printed and laser cut parts do not fit together, make the necessary changes to your files to account for the tolerances.

Laser cutting deliverables

- A properly formatted .pdf file, **on Canvas**, named: 'part2_<yourname>_ece115_lab1.pdf'

1.3.1.b.2 3D printing

In this part of the lab you will:

- Use your SolidWorks-generated file from the pre-lab exercise (Part #1; .stl file) to make your part from ABS plastic using the 3D printer in the lab

3D printing deliverables

- A working .stl file, **on Canvas**, named 'part1_<yourname>_ece115_lab1.stl'

1.3.2 System-level design: System Requirements, Block diagrams, and Design Specifications

This activity introduces the practice of system-level thinking and design, including developing and communicating complex systems using block diagrams.

In this part of the lab, you will:

- Practice the process of designing a system to meet system requirements
- Practice creating a block diagram of a system design
- Practice translating system requirements into design specifications

1.3.2.a Step 1: Understanding the problem and system requirements

Problem: The garage door is sometimes open when it should be shut, and nobody inside the house notices until a neighbor works up the resolve to bring the issue to someone's attention.

System requirements: **System requirements** capture the important facets of final system functionality that are necessary to meet the user's needs. Suppose the final system does not achieve these functional requirements. In that case, it will not be possible to validate the system during user testing, and the system will presumably fail to find user acceptance. One challenge of writing good system requirements is ensuring that the requirements do not lock the design into a specific solution.

The following is a partial/example list of system requirements for a garage door system that you will improve:

1.3.2.a.1 Garage door 'inappropriately-open' time <15 minutes

1.3.2.a.2 Garage door able to be kept open beyond 15 minutes, according to user needs

1.3.2.a.3 Easy to operate by non-expert users

1.3.2.a.4 …

Exercise: Develop at least one additional system requirement and one "optional" or "nice-to-have" constraint to add to the list above, based on your knowledge of the problem. Write the updated complete list of requirements **in your report.**

1.3.2.b Step 2: System design(s) and block diagram(s)

Now that you have a list of requirements, work with a partner to create a system design that will meet the requirements from 1.3.2.a.

Draw a functional block diagram for the system design, including labels for EVERY input/output/connection between blocks, as well as the block descriptions. Figure 3 includes an example that is at an appropriate level of detail for this exercise.

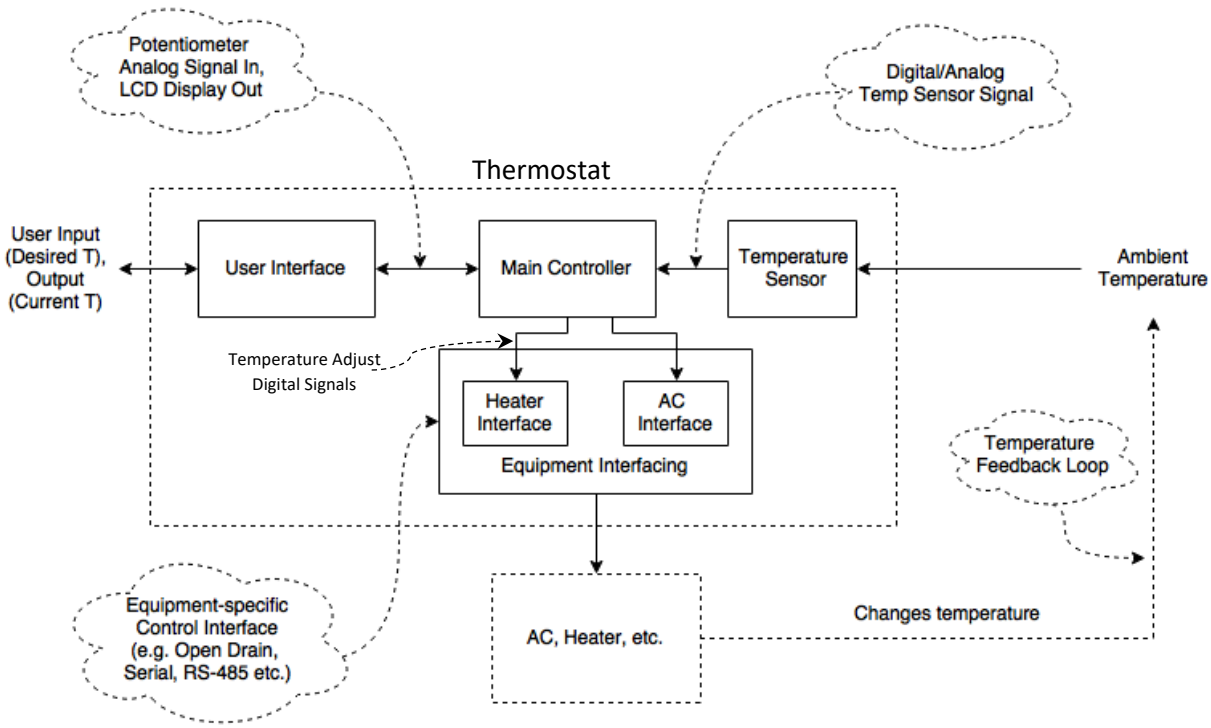


Figure 3: Example functional (system level) block diagram of a household thermostat. Key components: A suitable level of detail is provided without specifying exactly what will be used, and all inputs/outputs from each block are labelled and directions are shown to know what blocks are passing back and forth.

1.3.2.c Step 3: Translating system requirements into design specifications

Using the system functional block diagram that you created in 1.3.2.b, and the system requirements from 1.3.2.a, develop a set of design specifications for your system.

Design specifications are the measurable characteristics of the design/system, with associated acceptable/not acceptable value thresholds, that you will use to determine if the design is built and behaving as you intend (at both intermediate and final stages of development). Design specifications always have a clear metric, including units of measurement, which objectively determine if the prototype/design meets the specification. See Table 1.

TABLE 1 (Trace Matrix): TABLE RELATING HIGH-LEVEL SYSTEM REQUIREMENTS TO THEIR LOWER-LEVEL SPECIFICATIONS.

Specification number	Specification description	Test to perform	Relevant requirement	Specification [units]	Measured Value [units]
1.1	Ball propelling distance	Fire the ball across the playfield, measure its peak travel distance	1*	Greater than 50 cm	
…	…	…	…	…	…

* Note: 'Relevant requirements' refers to the requirements list in the Project Description

Note: While it is sometimes trivial to create a specification based on a requirement, the translation from system requirements into design specifications is frequently challenging, and it takes iteration, attention to detail, and collaboration to achieve an efficient and sufficient set of specifications.

To document your design specifications, create a 'trace matrix' that describes the specifications and traces them to their relevant requirement.

Regarding making specifications, a way to start this process is by making a list of all of the system functional block inputs and outputs and asking/answering: 'what are the units and approximate value ranges for each of the inputs/outputs?'

Typically the requirements and specifications are "living documents", and can be refined – often added to rather than reduced – as products get more refined and nuances reveal themselves. That is to say --- you always try to be the most complete with a list, until you realize later on that you forgot or missed a detail.

1.3.3 Pinball System Design Plan

The first step in creating a system is to make a plan that starts with the system requirements and works through to a refined block diagram with subsystem inputs and outputs. You will work with a partner of your choice for the remainder of the quarter.

- Understand the system requirements for the pinball machine
- Create a very high-level block diagram of the pinball system that can meet the system requirements. Details such as specific components used are not necessary at this stage.

This activity should help you plan ahead to understand how the pinball should be laid out, which is the next section.

1.3.4 Pinball Machine Playfield Mechanical Design Cycle

This activity of Module 1 applies the skills and practice from 1.3.2 and 1.3.1, with the pinball system design, to design the pinball machine playfield design while following a coherent system design cycle through Planning, Building, Testing, and Assessing.

You and your partner will create a new pinball machine playfield mechanical design, along with appropriate documentation of the following steps in this cycle, including associated system requirements, block diagram(s), design specifications, prototypes, test measurements, and verification/demonstration of specifications and requirements.

In Module 1, you will complete the first design cycle in which you will first sketch a 1-to-1 scale playfield design to understand how the parts can be made and then integrated. You will complete a second design cycle where you will mock it up, again 1-to-1 at scale, to integrate your learning from the first cycle to create a refined design. You will keep this for the rest of your class.

1.3.4.a Pinball Machine Playfield Mechanical Design Cycle - Sketch

- Sketch a scaled version of your anticipated playfield on two letter-sized sheets of paper (i.e. one top view, and one side view) with consideration of system requirements
    - Designate locations for each of your intended components including mechanical structures and moving components
    - Account for placement of motors, wiring
- Write the mechanical design specifications for your pinball machine's mechanical subsystem, showing that they meet all the requirements as outlined in the Pinball Machine Project Description page for the class.

Verify that your design, when built, will have met the requirements of the pinball machine. Show the teaching staff for feedback.

1.3.4.b Pinball Machine Playfield Mechanical Design Cycle - Mockup

- You will mock up the entire pinball machine using cardboard. This system should be mocked up 1-to-1.
- Designated locations for each of your intended components, including mechanical structures and moving components, should be tested in place with tape.

Place special emphasis on where components should go, how they should be hooked up (especially if they are providing movement). Show the teaching staff for feedback.

1.3.5 Pinball Plunger Testbed Mechanical Design Cycle

This activity applies the skills and practice from 1.3.1 and 1.3.2 to create and build the mechanical design for a pinball plunger. This first prototype should be constructed as a module that you can drop into your machine later on. This means, the plunger should have framing around it (e.g., a box-like frame) that allows it to be clamped down to a table and appropriately tested.

1.3.5.a Pinball Plunger Mechanical Design Cycle: Plan

- Create a specific/refined system and mechanical sub-system design that meets the relevant requirements and uses provided/available electrical and mechanical system components (you will receive a bag of parts in lab)
- Use the sub-system design to write mechanical design specifications for your plunger testbed mechanical design. These specifications/metrics will be the basis of your verification testing in section 1.3.5.c.
- Draw (using CAD tools) your plunger testbed mechanical design

1.3.5.b Pinball Plunger Mechanical Design Cycle: Build

- Create/modify drawings of parts and assemblies by hand or in SolidWorks.
- Fabricate parts
- Assemble the testbed
    - Include mechanical components, e.g., motors, springs, fasteners, etc.
- Document any designs/changes with drawings, files, or pictures, as appropriate

Important: the pinball plunger should have a housing around which you can appropriately test. This is typically some sort of box frame or U-channel out of wood or 3D printed. If you do not have a way to secure down the plunger and test its functionality, we will ask you to go back and include a sufficient housing before grading.

1.3.5.c Pinball Plunger Testbed Mechanical Design Cycle: Test

- Perform mechanical subsystem verification tests, based on specifications/plans in 1.3.5a, and record results in a trace matrix.
- To complete these tests.

1.3.5.d Pinball Plunger Testbed Mechanical Design Cycle: Assess

- Analyze verification test results
- Create notes for potential redesign (in lab report)

A Final Tip: If you designed your plunger to be modular, you can keep what you built here and fit it into your final pinball machine at the end. Consider this for every module --- Modules 1-3 will end with building one or several components of your pinball machine --- if you make it mechanically and electrically modular/contained, it will be "as easy as" drop into the playfield in the end.